

Week 8 Week 8 · Test and improve

· Paper completion route (no device) · paper route answers

Teacher only. Keep this sheet away from pupil copies.

P1 GRIDS - CORRECT CONTENT. 1 Player reset: when green flag clicked / go to x: -180 y: -120. 2 Player wall rule correct: when green flag clicked / forever / if < touching [Maze] ? > then / go to x: -180 y: -120. 3 Player wall rule Test Copy: when green flag clicked / if < touching [Maze] ? > then / go to x: -180 y: -120 - the forever is absent and must stay absent in the copy. 4 Player Hazard rule: when green flag clicked / forever / if < touching [Hazard] ? > then / go to x: -180 y: -120. 5 Player win correct: when green flag clicked / say [] / go to x: -180 y: -120 / wait until < touching [Finish] ? > / say [Made it!]. 6 Player win Test Copy: when green flag clicked / go to x: -180 y: -120 / wait until < touching [Finish] ? > / say [Made it!] - the empty say is absent. 7 right arrow: change x by 10. 8 left arrow: change x by -10. 9 up arrow: change y by 10. 10 down arrow correct: change y by -10. 11 down arrow Test Copy: change y by 10. 12 Hazard: when green flag clicked / go to x: 0 y: -100 / forever / glide 2 secs to x: 0 y: 100 / glide 2 secs to x: 0 y: -100. 13, 14, 15 Maze, Finish and Stage: no blocks.

P1 WORD BANK GATE. Every bank on every grid holds only these Week 8 blocks: when green flag clicked, when [arrow] key pressed, go to x y, change x by, change y by, glide secs to x y, forever, if then, wait until, touching [] ?, say []. There is no set block, no score, no lives and no counter block anywhere. If a pupil asks for one, the answer is: not in this unit. Note for staff: move () steps, point in direction () and say () for () seconds appear elsewhere in the unit but not in Week 8, so they are correctly absent from these banks.

P2 MAZE - CORRECT PLACEMENTS. S is at x -180, y -120. Clear space is x 30, y 0 and touches nothing. The wall 1 test point x -60, y 0 puts the counter edges inside inner wall 1, which runs x -70 to -50 and y -140 to 80. The wall 2 test point x 60, y 0 puts the counter edges inside inner wall 2, which runs x 50 to 70 and y -80 to 140. The Finish card covers x 168 to 192 and y 108 to 132, so a counter cross at x 180, y 120 overlaps it. The Hazard travels only the x 0 column.

P2 ROUTE - CORRECT ANSWER. Corners in order: x -180 y 100, then x 30 y 100, then x 30 y -120, then x 180 y -120, then x 180 y 120. Presses per leg: 22, 21, 22, 15, 24. Total 104 presses. Touching Finish at the end: yes. A correct drawing is five straight legs with square corners, no line crossing any grey rectangle, and the downward leg on the x 30 column rather than the x 0 column. If a pupil asks why not x 0: because the Hazard travels the x 0 column, and at x 30 the Player counter edge is 22 units clear of the widest part of the Hazard counter.

P3 TABLE 1 ANSWERS, from x -180 y -120 each time. right arrow: x -170, y -120. left arrow: x -190, y -120. up arrow: x -180, y -110. down arrow on a correct game: x -180, y -130. Correct if the pupil resets before every key. A pupil who does not reset will produce a drifting position; that is a genuine wrong result, so ask them to trace it again with the reset.

P3 TABLE 2 ANSWERS. Row 1, the hat block, moves nothing: x -180, y -120. Row 2 copies change y by 10 and gives x -180, y -110. Traced result: x -180, y -110.

Expected: x -180, y -130. Same or different: different. Teaching point worth naming aloud - the down key on the Test Copy puts the counter in exactly the same place as the up key. That is a very clear way to see the fault without any jargon.

P3 TABLE 3 ANSWERS. Flag row: touching Maze, no; x -180, y -120. Row 1: the if is asked once and the answer is no, so nothing happens; position unchanged. Row 2: is there a forever above the if - no. Row 3: the counter is now at x -60, y 0 and touching Maze is yes. Row 4: does the wall check run again - no, because there is no forever. Row 5: the counter cross is still at x -60, y 0, inside the wall. On a correct game rows 4 and 5 read yes, and x -180, y -120.

P3 TABLE 4 ANSWERS. Hazard y in order: -100, -50, 0, 50, 100, 50, 0, -50, -100. Hazard x is 0 on every row. Overlapping: no on every row. The Player counter at x 30 has its left edge at x 22; the Hazard counter reaches x 10 at its widest. They never meet. When the pupil then places the Player cross on the Hazard cross, the Hazard contact rule sends the Player counter to x -180, y -120. The Hazard counter keeps going - the contact rule belongs to Player and does not move Hazard. That is the Week 7 bug the pupil already met; it is correct in all three Week 8 files.

P3 TABLE 6 AND 7 ANSWERS FOR THE TEST COPY. Check 1 Arrows: fault. Right, left and up pass. Down gives x -180 y -110 instead of x -180 y -130. Cause: change y by 10 on the down key script. Repair: change y by -10. Re-check: x -180, y -130. Check 2 Walls: fault. Clear space passes. Both wall contacts leave the counter in the wall. Cause: no forever, so the check runs once at the flag when Player is not touching a wall. Repair: put the if inside a forever under the flag. Re-check: both wall contacts return the counter to x -180, y -120. Check 3 Finish: PASS. No early win. One Made it! card when the counter reaches Finish. Staying there adds no second card. Check 4 Feature: PASS. The Hazard repeats its journey, avoidance at x 30 leaves the Player counter free, and contact returns it to the safe start. Check 5 Restart: fault, on two counts. The old Made it! card stays on the maze, and the wall still does not return the counter. Positions do reset and a new win does happen. Cause of the card: no empty say block at the top of the win script. Repair: add say [] as the first block under the flag. Re-check: the card comes off at the flag.

P5 ORDER OF REPAIR. One cause at a time, then the same check again, then check 3 and check 4 again to show nothing that passed has broken. The pack models fault A in full at I Do 2; faults B and C are found and explained but the shipped Test Copy is never repaired, so pupils repair them on their own paper copy only.

ROUTE CONSEQUENCE OF FAULT A, if a pupil asks. On the Test Copy, leg 3 of the ordinary route is 22 down presses. They send the counter up instead of down. Scratch would hold the sprite at the top of the stage. On paper, stop the counter at the top edge of the maze and record: the counter went up and stopped at the top. That is a correct actual result. Do not spend lesson time on the edge-holding rule.

WHAT CORRECT LOOKS LIKE WHEN A PUPIL POINTS. Check 1 is correct if the counter cross lands on the right grid intersection and the pupil can point to the block whose number caused it. Saying ten or minus ten aloud is correct. Writing is not required. A trace table is correct if the pupil points to each row in order while the counter sits on the matching coordinate; a scribe writes the pupil's exact words and marks them as scribed.

WHAT CORRECT LOOKS LIKE FOR FINDING A FAULT. Correct if the pupil puts a finger on the change y by 10 line, or on the gap where the forever should be, or on the top of the win stack where the empty say should be. The pupil does not have to use the

words fault, forever or expected. Pointing at the right line, after tracing, is the demonstration.

WHAT CORRECT LOOKS LIKE FOR A REPAIR. Correct if the pupil says make it minus ten, or draws a minus sign in front of the 10, or points to change y by -10 in the word bank, or asks a scribe to write the new line. Writing the line is one way and not the only way. What must be observed is that the pupil changed one thing and then traced the same check again.

WHAT CORRECT LOOKS LIKE FOR EXPECTED AND ACTUAL. Correct if the pupil gives the expected result before the trace and the traced result after, even in three words each. Should go down. Went up. is a complete pass. It works is not, and neither is it is broken - send the pupil back to the setup, the action, the expected and the actual, exactly as the Starter teaches.

WHAT CORRECT LOOKS LIKE FOR AN ALL-PASS. Correct if the pupil says or writes all pass and names one improvement. Write the keys on the page so people know is a good improvement and needs no code change. Record it in the pupil's own words. Do not prompt for a fault that is not there.

WHAT CORRECT LOOKS LIKE FOR THE EMPTY-SCRIPT GRIDS. Maze, Finish and Stage are correct if the pupil writes no blocks, says no blocks, draws a line through the empty grid, or is asked does this one have any blocks and shakes their head after looking. Checking that a sprite has no code is a real check and is credited.

WHAT CORRECT LOOKS LIKE FOR THE TESTER STEP. Correct if the tester traced the pupil's script from the start the pupil named, and the pupil wrote or dictated the tester's exact words. If the tester's traced result differs from the pupil's, that is a finding, not a failure: trace it together and record which of the two followed the block as written.

EVIDENCE ROUTING. Complete the teacher guide Individual evidence table and the six-row final evidence check from your own observation of the pupil. Record the paper route in the teacher guide only. The AQA summary sheet is completed by the teacher exactly as on the screen route, and no awarding-body document is altered, annotated or written on by a pupil.

What this week evidences

AQA 71638 Programming with Scratch (unit 6), Level One, outcome 6: "Tested the game." The pack's own teaching wording is: "Observe the pupil testing the game, recording expected and actual results, and retesting after a repair." The pack also states, twice, that "AQA summary sheets remain the official evidence" and "Record completed outcomes on the AQA summary sheet. Do not infer creation from supplied code." The five-check record, the written expected and actual notes and the improvement are named in the pack as local teaching scaffolds: "outcome 6 asks for testing and does not require finding a fault", and "An all-pass result is valid." Week 8 also carries the whole-unit review: outcome 6 is claimed here and outcomes 1 to 5 are reviewed individually using Planning/GROW_Computing_Final_Evidence_Check.docx. "A partly finished game gets a recorded next step, not an assumed completion."

Known limit of the paper route

Outcome 6 is met on paper. Three limits are real, and are named here rather than hidden. FIRST, a paper trace samples time; Scratch runs it continuously. Check 4 and

the Challenge edge case "contact during a glide" are traced at half-second steps, so the Hazard is seen at y -100, -50, 0, 50, 100 and back. A contact that would happen between two of those instants is not seen on paper. The check still evidences expected against actual and a retest, but it evidences a sampled journey, not a continuous one. Use that wording in the teacher guide. SECOND, paper can only find a fault that is already visible in the written script. On a device the machine is an independent authority and will surface faults nobody wrote down. On paper the pupil executes only what is on the page, so a mismatch between what the pupil wrote and what the pupil meant is findable, and a mismatch between what the pupil wrote and what Scratch would actually do with it is not. This is exactly why the Test Copy is supplied with three faults already written down: it gives the paper route something real to find and repair. THIRD, the Exit save-and-reopen loop has no paper equivalent, because no .sb3 file exists on this route. The substitute is: the pupil dates and signs the pages, puts them in the folder, the folder is closed, and the pupil reopens it and traces one named check from the pages alone. That evidences that the record is complete enough to work from later. It does not evidence a saved digital file, and the teacher guide should say so in plain words rather than record a filename. ONE THING THAT IS NOT A GAP. If the pupil's own game passes all five checks, the repair and the retest are demonstrated on the Test Copy rather than on their own game. That is identical on the screen route, where an all-pass pupil also repairs nothing of their own. The pack states that outcome 6 asks for testing and does not require finding a fault, and that an all-pass result is valid. ONE RISK THAT MUST BE MANAGED, NOT WRITTEN AWAY. The trace is only evidence if it is faithful. A pupil who moves the counter the way the block was meant has restated an intention, not tested a script. Observation must therefore be closer on paper than on screen: the adult reads the block and never the direction, and watches the counter move. If the adult supplies the direction, the outcome is not evidenced and must be left pending with a recorded next step.